map-charting-student-math-misunderstandings

MAP2025_Private&Public 2nd

Rank	2
Team	looking for remote job
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Topic ID	612145
Version	Original

Source: https://www.kaggle.com/competitions/map-charting-student-math-misunderstandings/discussion/612145
Retrieved with Kaggle CLI on 2026-10-21.

Thanks so much for the host and I am very delighted to join the competition. My teammate Baiph was very hard working and we did the same contribution to the final results, both of us have the different training pipeline in that way we can be 2nd in public and private leaderboard.

P.S. I am looking for a remote job in NLP or LLM area, perefered UTC +8 time zone, please email me if it's possible.

Overview of our solution

The task of this competation is to predict the misunderstandings according to the student's explaination. This is a very easy classification problem but the metric of this comptation is MAP@3 which is used in Ranking Problem normally, so I think the label is ambiguous sometimes, the student's explainations can be labeled in more than one target.

Based on above analysis, what we need to do is denoise or give the student's explaination some soft labels according to oof in case of label leakage.

As for data, there are over 30 labels had limited data, so I used many commerical LLM to generate external data.

As for model selection, we used LLM model as the backbone we tried qwne3, qwen2.5, mistral, phi-4.

As for training strategy, we all convert the classification problem as the multi-choices generation problem, only in this way, we can use vllm to finish the inference timely.

HZM Part

I am here to introduce my best private sigle model qwen3 -14b which can be much more easy to understand. The training pipeline can be divided into 4 steps.

Step1: Generate the ext data (both I and Baiph used them)

Generate one student output with a rare candidate for each question.
Incorporate contrastive learning with the labels.

```
For example:
 I want to generate student
```

I used gpt-4, gpt-5, claude-sonnet, gemini, seed, doubao, and generated 80K data.

Step2: Train LLM to label ext data

Train 4 LLM models on all train.csv and then label ext data (use this as soft label in step4 training), which were phi_4_reasoning_14b,qwen3_32b,mistral_12b,qwen2_5_72b

```
The following  training prompt:



```

Step3: Train LLM in 5 fold to generate soft label

Train 4 LLM models in 5 fold, avg the oof.csv (use this as soft label in step4 training)

Step4: Multi-loss desgin for qwen3-14b

I changed the SFTTrainer and combined the generation loss (hard label) and classification loss (soft label) target_token_id meant that the each option will convert to a token id in vocab.

```
 ():
     ():

        target_token_id = .model.config.target_token_id
        target_token_id = torch.tensor(target_token_id, device=model.device)

        labels = inputs[]
        mask = torch.isin(labels, target_token_id)
        labels[~mask] = -
        inputs[] = labels
        _, outputs = ().compute_loss(model, inputs, return_outputs=,
num_items_in_batch=num_items_in_batch)
        logits = outputs.logits
        loss = outputs.loss

        shift_logits = logits[..., :-, :].contiguous()
        shift_labels = labels[..., :].contiguous()

        logits_target = []
         i  ((shift_labels)):
            lbl = shift_labels[i].cpu().numpy()
            target_idx = np.where(lbl != -)[][-]
            logits_target.append(shift_logits[i][target_idx][target_token_id])


        logits_target = torch.stack(logits_target, dim=)

        labels_target = inputs[].to(outputs.logits.device)
        soft_loss = F.cross_entropy(logits_target, labels_target)


        loss = loss +  soft_loss
         (loss, outputs)  return_outputs  loss
```

After the above steps the CV on fold-0 can achieve** 0.955** and public is **0.950** private is **0.946**

Baiph Part

1、Misunderstandings expansion

Convert 65 categories into 37 categories and use DeepSeek to expand and explain brief misunderstandings

```
mis2reason = {
        : ,
        : ,
        : ,
```

```
(A) \( \frac{}{} \) (B) \( \frac{}{} \) (C) \( \frac{}{} \) (D) \( \frac{}{} \)

A. \( \frac{}{} \)



I think that /  the answer,  it
```

3、Training strategy

LR=2e-4，BS=4*2 Epoch=2

Additional data combined with distillation can enhance LB and PB, but there is no benefit after fusion

model	data	LB	PB	use final	infere time
Qwen25-14B-AWQ	total data	0.948	0.946	submission 1（best）	20min
Qwen3-14B-AWQ	fold1	0.948	0.943	submission 1（best）	20min
Qwen25-32B-AWQ	fold1	0.948	0.944	submission 1（best）	40min
Qwen25-32B-AWQ	total data	0.949	0.945	submission 1（best）	40min
QWQ-32B-AWQ	ext data 80k total data training distal	0.949	0.948	submission 2	40min
Qwen25-32B-AWQ	ext data 80k fold1 distal	0.949	0.947	submission 2	40min
Qwen3-14B-AWQ	ext data 80k fold1 distal	0.950	0.945	submission 2	20min

4、Ensemble

- Four 37-classification models
- Four 65-classification models
- Equal-weight average fusion

Supplemental Q&A

> **Levin · 2025-10-17T06:37:32.593000**
>
> Brilliant write-up—your data generation pipeline and overall architecture are impressively crafted.
> On the note that “we converted classification to multi-choice generation to make vLLM inference timely,” I found a workable alternative that keeps it as classification: merge LoRA → split base & classifier head → GPTQ-4bit the base only → run vLLM in embed mode (LAST pooling, no normalization) → feed embeddings into an external FP16 linear head for logits. It gave me fast, stable inference without switching to generation.

> **Bilzard · 2025-10-17T06:49:45.993000**

Congratulations for winning 2nd.

@leehann I'm interested in how to utilize contrastive examples.
In the following explanation, what `False_Misconception:Additive_same_math_problem`, `False_Misconception:Additive_different_math_problem`, etc. mean?
Are they in-context example of student's explanation sampled from the following conditions?

1. misconception=`False_Misconception:Additive`, QuestionId={different}
2. misconception=`False_Misconception:Additive`, QuestionId={same}
3. misconception=`True_Misconception:Additive`, QuestionId={different}
4. misconception=`True_Misconception:Additive`, QuestionId={same}

And do you mind sharing example prompt?

```
For example:
 I want to  student's explation which is True_Misconception:Additive(  as hard label in step4
training),
I will
False_Misconception:Additive_same_math_problem
False_Misconception:Additive_different_math_problem
+
True_Misconception:Additive_same_math_problem  True_Misconception:Additive_different_math_problem
put the contrastive examples  the ChatGPT  Claude  better data
```

HZM · 2025-10-17T06:50:38.340000

that's so nice job you have done, i tried it before but did not work. hope you can share some code

HZM · 2025-10-17T06:52:52.610000

you are right, i sampled the examples with problem id, try to make sure the generated data was more reliable
check the following prompt

```
instruct_template =
```

Bilzard · 2025-10-17T07:10:58.023000

Sorry, I get it. I guess they are equally sampled to make sure examples from each four categories shown above. (Because QuestionText is shown each example, the LLMs can tell each examples from what question.)